Local asymptotic stability of rational polynomial ODEs is undecidable

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Abstract

No Turing algorithm can decide local asymptotic stability of the origin for every exactly encoded rational polynomial autonomous ODE, when dimension and degree are unrestricted. This follows from the recent global asymptotic-stability theorem of Shevchenko, Liu and Liu [1]. We give the reduction for the precise formulation of AIM Problem 338, including forward existence near the equilibrium. For a homogeneous polynomial vector field, local asymptotic stability implies global asymptotic stability by rescaling both the initial condition and time.

1 The question and the cited theorem

The question is whether exact polynomial coefficients suffice for a terminating decision procedure for local asymptotic stability. The distinction between stability and attraction matters: a stable equilibrium whose nearby solutions do not converge to it is a negative instance. AIM Problem 338 [2] asks this question for rational polynomial continuous-time systems without a bound on dimension or degree.

Shevchenko, Liu and Liu [1] prove undecidability of global asymptotic stability even for a suitable fixed dimension and fixed homogeneous degree. This note records the consequence for the full local question. We use Theorem 1.5 of their October 8, 2026 preprint, version 1; the undecidability construction is theirs. The additional argument here is the elementary local-to-global implication for homogeneous fields and the exact conversion of the decision problems.

An input consists of an integer $n \geq 1$ and a finite monomial description of

$$f \in \mathbb{Q}[x_1, \ldots, x_n]^n, \qquad f(0) = 0, \qquad \dot{x} = f(x) \quad \text{on } \mathbb{R}^n.$$

Coefficients are exact rationals and exponents are nonnegative integers. Write $\phi_f(t, x_0)$ for the unique maximal classical solution from x_0, and let $\|\cdot\|$ be the Euclidean norm. We use precisely the following meaning of $\mathrm{LAS}(f)$:

(i) For every $\varepsilon > 0$ there is a $\delta > 0$ such that, if $\|x_0\| < \delta$, the solution exists for all $t \geq 0$ and $\|\phi_f(t, x_0)\| < \varepsilon$ for all $t \geq 0$.

(ii) There is an $r > 0$ such that, if $\|x_0\| < r$, the solution exists for all $t \geq 0$ and $\phi_f(t, x_0) \to 0$ as $t \to \infty$.

Thus forward completeness is required near zero; it is not assumed for arbitrary initial states. A decider must halt on every valid finite input and return the truth value of this conjunction.

The predicate GAS(f) requires $f(0) = 0$, condition (i), and existence for all $t \geq 0$ with convergence to zero for every $x_0 \in \mathbb{R}^n$. This agrees with the convention in [1, §1.1].

For fixed positive integers N, d, let HGAS(N, d) be the set of rational coefficient codes for homogeneous degree-d polynomial vector fields on $\mathbb{R}^N$ satisfying GAS. Invalid codes are excluded. A set is recursively enumerable if a Turing machine halts precisely on its members. It is many-one complete if it is recursively enumerable and, for each recursively enumerable set A, there is a total computable function q such that $e \in A$ if and only if $q(e)$ belongs to that set.

Theorem 1 (Shevchenko–Liu–Liu [1, Theorem 1.5, p. 4])**.** *There are a fixed integer $N_* \geq 1$ and a fixed odd integer $d_* \geq 3$ for which* HGAS(N_*, d_*) *is many-one complete.*

The coefficients and the autonomous continuous-time model in this theorem are the same as those above. Its proof is in [1, §15, pp. 44–45].

2 From local to global stability

Homogeneity allows any initial state to be moved into the local attraction ball. The corresponding change of time preserves the entire forward half-line.

Lemma 2. *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a homogeneous polynomial vector field of degree $d \geq 1$. Then* LAS(f) *holds if and only if* GAS(f) *holds, with the definitions above.*

Proof. Global asymptotic stability implies both local conditions, with any positive attraction radius. Conversely, assume LAS(f) and let $r > 0$ be supplied by condition (ii). Condition (i) already gives the stability required for GAS.

Fix $x_0 \neq 0$ and choose

$$\lambda = \min\left\{1, \frac{r}{2\|x_0\|}\right\} > 0.$$

Then $\|\lambda x_0\| \leq r/2 < r$, so $y(t) = \phi_f(t, \lambda x_0)$ exists for every $t \geq 0$ and tends to zero. Define, on the whole half-line,

$$x(t) = \lambda^{-1} y(\lambda^{1-d} t).$$

The chain rule and $f(\lambda x) = \lambda^d f(x)$ give

$$x'(t) = \lambda^{-d} f\big(y(\lambda^{1-d}t)\big) = \lambda^{-d} f\big(\lambda x(t)\big) = f(x(t)), \qquad x(0) = x_0.$$

Since λ^{1-d} is a fixed positive number, x is defined for all nonnegative times and $x(t) \to 0$. By uniqueness it agrees with the maximal solution from x_0 on their common domain. This global continuation rules out a finite forward endpoint of that maximal solution.

For $x_0 = 0$ the solution is identically zero, since $f(0) = 0$. Hence every initial state has a forward-complete solution converging to zero. Together with condition (i), this proves GAS(f). □

3 The decision problem

Corollary 3. *There is no Turing algorithm that halts on every finite exact rational polynomial input of unrestricted dimension and degree with $f(0) = 0$ and correctly decides conditions (i) and (ii). In particular, the answer to AIM Problem 338 at the revision in [2] is negative.*

Proof. Suppose such a decider D exists. Fix N_*, d_* from Theorem 1. We construct a total decider C for HGAS(N_*, d_*).

On an arbitrary source code, first check the finite rational coefficient syntax and the required dimension and degree. Combine any repeated monomials by exact rational addition, then verify that each nonzero monomial has total degree d_*. Zero coordinate polynomials are allowed. The checks terminate; on an invalid code, return false, as required by the definition of HGAS. In a format that stores only coefficients, the fixed dimension and degree are hardcoded. If they are explicit metadata, check them as well.

On a valid code, write the same vector field as the finite monomial lists accepted by D. A dense degree-d_* coefficient list can be converted by enumerating its finitely many monomials and omitting zero coefficients. No coefficient is approximated. As $d_* \geq 1$, the resulting field vanishes at zero. Thus the input to D is valid and D halts. Lemma 2 identifies its answer about LAS with the required answer about GAS. The resulting C is total. The numerical values of the fixed existential constants need not be extracted: their existence suffices to give a finite program with those constants hardcoded.

Let K be the set of codes of pairs (M, w) for which the Turing machine M halts on w. It is recursively enumerable by simulation. It is undecidable: given a hypothetical total halting decider H, a machine B on a machine code u could loop if $H(u, u)$ answers yes and halt if it answers no. On its own code b, either answer $H(b, b)$ would then be wrong.

By Theorem 1, a total computable map q reduces K to HGAS(N_*, d_*). Running q and then C would therefore decide K on every input, a contradiction. □

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References

[1] I. O. Shevchenko, J. Liu and X. Liu, *Global asymptotic stability of homogeneous polynomial vector fields is undecidable*, preprint, arXiv:2610.12434v1 [math.DS], October 8, 2026. arXiv:2610.12434v1.

[2] AIM problem catalogue, *338: Decidability of local asymptotic stability for polynomial ODEs*, revision `c9929805cece044705abaaebe1cebddac1ba11a6`. Immutable problem statement.